Arduino code

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include <math.h>


// Define constants

#define SERIAL_BAUD_RATE 9600

#define MAX_DISTANCE_CM 400

#define MIN_DISTANCE_CM 0

#define ANGLE_STEP_DEGREES 30


// Define structure for radar data

Typedef struct {

  Int angle;

  Int distance;

} RadarData;


// Function to process serial data

Void processSerialData(char *data, RadarData *radarData) {

  // Split data into angle and distance

  Char *angleStr = strtok(data, “,”);

  Char *distanceStr = strtok(NULL, “.”);


  // Convert angle and distance to integers

  radarData->angle = atoi(angleStr);
```

```
  radarData->distance = atoi(distanceStr);

  // Validate angle and distance
  If (radarData->angle < 0 || radarData->angle > 180) {
    radarData->angle = 0;
  }
  If (radarData->distance < MIN_DISTANCE_CM || radarData->distance >
MAX_DISTANCE_CM) {
    radarData->distance = 0;
  }
}

// Function to update radar system
Void updateRadarSystem(RadarData *radarData) {
  // Clear background (simulated)
  Printf("Clearing background...\n");

  // Draw radar system (simulated)
  Printf("Drawing radar system...\n");
  Printf("Angle: %d°\n", radarData->angle);
  Printf("Distance: %d cm\n", radarData->distance);
}

Int main() {
  // Initialize serial communication (simulated)
  Printf("Initializing serial communication...\n");
```

```
    // Create radar data structure
    RadarData radarData;

    // Simulate serial data reception
    Char serialData[] = “90,200.”;

    // Process serial data
    processSerialData(serialData, &radarData);

    // Update radar system
    updateRadarSystem(&radarData);

    return 0;
}
```